

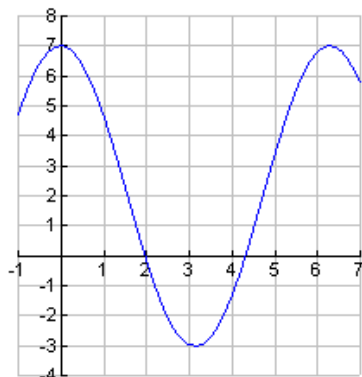
AP Precalculus
Unit 2 Wkst A

Name: _____

A) For each of the following graphs, write the equation of the midline and find the amplitude.

B) Then write an equation in the form $y = a \sin(x) + d$ or $y = a \cos(x) + d$

1.

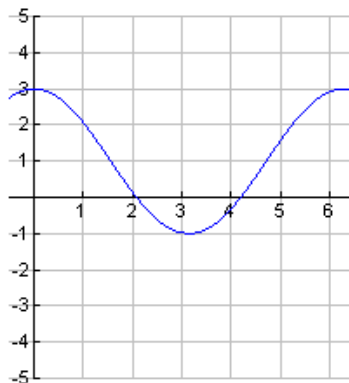


Midline: _____ d = _____

Amplitude: _____ a = _____

Equation: _____

2.

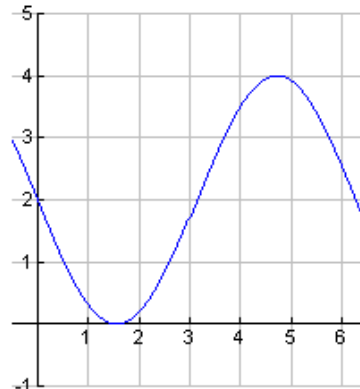


Midline: _____ d = _____

Amplitude: _____ a = _____

Equation: _____

3.



Midline: _____ d = _____

Amplitude: _____ a = _____

Equation: _____

For each of the following functions find the (a) midline, (b) amplitude, (c) length of the period, and (d) boundaries of one period. Then (e) graph one period of the function on graph paper.

4. $y = 4 \cos\left(x + \frac{\pi}{4}\right) - 1$

6. $y = 3 \sin\left(2x - \frac{\pi}{2}\right) + 2$

5. $y = -\sin\left(\frac{2\pi}{3}x\right) + 5$

7. $y = 3 \cos(2x - \pi) - 2$

Describe the **transformation** (translation/shift, dilation/stretch-shrink, reflection) of $g(x)$ compared to $f(x)$. Be specific. Make sure to tell direction and amount of the transformation.

8. $f(x) = \sin x$
 $g(x) = 5 \sin(x - \pi)$

10. $f(x) = \sin x$
 $g(x) = \sin 4x$

9. $f(x) = \cos x$
 $g(x) = -\cos x - 3$

11. $f(x) = \cos x$
 $g(x) = 8 - \cos x$